Revising the Dust Growth Time in the 3-Scalar Dust Model

April 15, 2026

1 Current Implemented Model

The current accretion block in `src/pgen/cgm_cooling_flow_full.cpp` (lines 1017–1038 as of 2026-04-15) uses

$$\tau_{\rm acc} = 150\,{\rm Myr}\left(\frac{100}{n_{\rm H}}\right)\left(\frac{50\,{\rm K}}{T}\right)^{1/2}\left(\frac{Z_\odot}{Z_{\rm gas}}\right)\left[1-\frac{Z_{\rm gas}}{Z_{\rm gas}+D_{\rm tot}}\right]^{-1}, \tag{1}$$

followed by a bin-wise accretion update

$$\dot{\rho}_{{\rm D},j} = \frac{\rho_{{\rm D},j}}{\tau_{\rm acc}\,a_j}. \tag{2}$$

Here $Z_{\rm gas}$ is the gas-phase metallicity tracer (`scalar00`), $D_{\rm tot} = D_{\rm s} + D_{\rm l}$ is the total dust-to-gas ratio carried by `scalar01` and `scalar02`, and a_j is the grain size assigned to the small or large bin. In other words, the current implementation:

1. uses the full gas-phase metal tracer as the accretion reservoir;
2. accretes directly onto the two dust scalars;
3. makes the growth rate scale as a_j^{-1}.

The effective e-folding time of dust growth in bin j is therefore

$$t_{{\rm grow},j} = \tau_{\rm acc}\,a_j. \tag{3}$$

For the small-grain size currently under discussion, $a_{\rm gr,s} = 3\times10^{-4}\,\mu{\rm m}$, and for a solar-abundance cold phase with

$$Z_{\rm gas} = 0.02, \qquad D_{\rm tot} = 0.004, \qquad Z_\odot = 0.02,$$

equation (3) becomes

$$t_{{\rm grow},s} = 0.27\,{\rm Myr}\left(\frac{100}{n_{\rm H}}\right)\left(\frac{50\,{\rm K}}{T}\right)^{1/2} = 190.9\,{\rm Myr}\,\frac{\sqrt{T}}{P/k_{\rm B}}, \tag{4}$$

where $P/k_{\rm B} = n_{\rm H}T$ is in ${\rm K\,cm^{-3}}$.

At

$$T = 100\,{\rm K}, \qquad P/k_{\rm B} = 10^6\,{\rm K\,cm^{-3}}$$

the current model gives

$$t_{{\rm grow},s} \simeq 1.9\times10^{-3}\,{\rm Myr}, \tag{5}$$

2 Comparison to Dubois et al. (2024)

Dubois et al. [1] use a different gas-accretion framework. Their section "Gas accretion" writes the growth rate as

$$\dot{M}_{\mathrm{acc},i,j} = \left(1 - \frac{M_{\mathrm{D},i,j}}{M_{Z,i}}\right) \frac{M_{\mathrm{D},i,j}}{t_{\mathrm{acc},i,j}}, \tag{6}$$

with an accretion time

$$t_{\mathrm{acc},i,j}^{-1} = \alpha \frac{\pi \bar{a}_j^2}{f_X m_{\mathrm{gr},i,j}} \rho_X u_{\mathrm{th},X}, \tag{7}$$

and, after simplification,

$$t_{\mathrm{acc},i,j} = 0.28\,\mathrm{Myr}\,\alpha^{-1} a_{0.005,j} s_{3,i} \frac{A_X^{1/2} f_X}{E_i(a_j) Z_X} \left(\frac{n}{1\,\mathrm{H\,cm^{-3}}}\right)^{-1} \left(\frac{T}{50\,\mathrm{K}}\right)^{-1/2}. \tag{8}$$

Several differences from the current AthenaK implementation matter:

1. Dubois do not accrete from a generic "all metals" pool. They accrete from the *limiting element* of the grain species, with explicit key-element limitation for silicates.
2. Their resolved formula is normalized to $a/5\,\mathrm{nm}$, not to sub-nm grains.
3. They include a temperature-dependent sticking coefficient and discuss the Le Bourlot form

$$\alpha_{\mathrm{LB}}(T) = \left(1 + 10^{-4} T^{3/2}\right)^{-1} \tag{9}$$

   in their appendix on sticking coefficients.
4. Their unresolved turbulence-driven boost is not active everywhere: it is only used for cells with $n > 0.1\,\mathrm{H\,cm^{-3}}$, $T < 10^4\,\mathrm{K}$, and a Jeans-length criterion.
5. Even in the subgrid model, accretion is truncated above $n_{\mathrm{max}} = 10^4\,\mathrm{H\,cm^{-3}}$. Dubois also note that water-ice mantles begin to matter already for $n \gtrsim 10^3\,\mathrm{H\,cm^{-3}}$.
6. Dubois further assume that gas-phase carbon is fully molecular above $n > 10^3\,\mathrm{H\,cm^{-3}}$, which suppresses carbonaceous growth in very dense gas.

Those points matter for the present problem. The current AthenaK model is allowed to operate on the full gas-phase metal pool, at arbitrarily high density, with a very small small-grain size, and without a temperature-dependent sticking suppression.

2.1 Draine-Style Physical Motivation

The same basic physical objection appears in standard interstellar-dust treatments. Draine-style dust models do not treat dust as a single undifferentiated "metal" fluid. They instead distinguish at least carbonaceous and silicate material, with separate compositions, size distributions, and limiting elements; see, for example, Weingartner & Draine [2] and Draine & Li [3]. In that language, `scalar00` in the present 3-scalar model should not be read as "the entire refractory reservoir available for growth." It is a mixed gas-phase metal tracer, only part of which should be accretable onto dust at any given time.

That point motivates the main structural change below: use a refractory subset of `scalar00` as the growth reservoir, rather than the full gas-phase metallicity tracer.

3 Proposed Revised $t_{\rm grow}$ Model

Within the existing 3-scalar framework, a conservative revised growth time can be written as

$$t^{\rm new}_{{\rm grow},j} = \begin{cases} 150\,{\rm Myr}\left(\dfrac{100}{n_{\rm H}}\right)\left(\dfrac{50\,{\rm K}}{T}\right)^{1/2}\left(\dfrac{Z_\odot}{Z_{\rm acc}}\right) & T < 10^4\,{\rm K},\ 0.1 \le n_{\rm H} \le 10^3\,{\rm cm}^{-3}, \\ \qquad \times \left(\dfrac{a_{{\rm eff},j}}{1\,\mu{\rm m}}\right)\alpha_{\rm LB}(T)^{-1}, & \\ \infty, & \text{otherwise}, \end{cases} \tag{10}$$

with

$$Z_{\rm acc} = f_{\rm ref}\, Z_{\rm gas}, \tag{11}$$

and effective accretion sizes

$$a_{{\rm eff},s} = 3\,{\rm nm} = 0.003\,\mu{\rm m}, \qquad a_{{\rm eff},l} = 0.1\,\mu{\rm m}. \tag{12}$$

The reasoning behind each new choice is direct.

1. **Use $a_{{\rm eff},s} = 3\,{\rm nm}$.** Dubois normalize their resolved accretion law to $a/5\,{\rm nm}$ in equation (8), but for the present note we shift the effective small-grain size slightly downward so the small bin remains in a more clearly PAH-like / ultrasmall-carbonaceous regime. This is still far less aggressive than 0.3 nm while staying close to the Dubois small-grain normalization.

2. **Use $a_{{\rm eff},l} = 0.1\,\mu{\rm m}$.** Dubois explicitly discuss $0.1\,\mu{\rm m}$ large carbonaceous grains in the cold neutral medium. It is the natural representative size for the large bin.

3. **Use $\alpha_{\rm LB}(T)$.** Dubois use a constant $\alpha = 1/3$ only in their unresolved turbulence model. Otherwise they revert to the Le Bourlot temperature-dependent sticking coefficient. In a simplified 3-scalar model without the Dubois turbulence PDF machinery, the resolved-regime sticking law is the cleaner default choice.

4. **Restrict growth to $T < 10^4\,{\rm K}$ and $0.1 \le n_{\rm H} \le 10^3\,{\rm cm}^{-3}$.** The lower-density and temperature thresholds come directly from the activation conditions of the Dubois unresolved model. The upper-density cutoff is chosen more conservatively than their formal $n_{\rm max} = 10^4\,{\rm cm}^{-3}$ because Dubois already note that ice-mantle and CO effects become important around $10^3\,{\rm cm}^{-3}$, and the present 3-scalar model does not track those dense-gas chemical restrictions.

5. **Use $Z_{\rm acc} = f_{\rm ref} Z_{\rm gas}$ rather than $Z_{\rm gas}$.** This is the 3-scalar proxy for the key-element limitation in Dubois and for the carbonaceous/silicate picture used by Draine-style dust models. It is not a claim that the current model has species-level fidelity; it is an explicit acknowledgement that only a subset of `scalar00` should be considered refractory and accretable.

For the numerical examples below, we adopt

$$f_{\rm ref} = 0.1,$$

as a deliberately conservative proxy for the unresolved refractory subset of the gas-phase metal tracer.

Condition	Current $t_{\mathrm{grow},s}$	Dubois et al. (2024)	Proposed $t^{\mathrm{new}}_{\mathrm{grow},s}$
$T = 100\,\mathrm{K}$, $P/k_{\mathrm{B}} = 10^6$	$1.9 \times 10^{-3}\,\mathrm{Myr}$	Dense-gas growth is curtailed: $n_{\mathrm{max}} = 10^4\,\mathrm{cm}^{-3}$, while ice-mantle and CO arguments already matter near $10^3\,\mathrm{cm}^{-3}$	off ($n_{\mathrm{H}} > 10^3$)
$T = 100\,\mathrm{K}$, $n_{\mathrm{H}} = 30\,\mathrm{cm}^{-3}$	$0.64\,\mathrm{Myr}$	Dubois quote $t_{\mathrm{acc,Sil}} \simeq 1.8\,\mathrm{Myr}$ and $t_{\mathrm{acc,C}} \simeq 9.5\,\mathrm{Myr}$ at solar abundance for their fiducial turbulent ISM case	$11.7\,\mathrm{Myr}$
$T = 8000\,\mathrm{K}$, $n_{\mathrm{H}} = 0.1\,\mathrm{cm}^{-3}$	$21.3\,\mathrm{Myr}$	Resolved accretion only, with $\alpha_{\mathrm{LB}} \ll 1$; growth is far slower than in the CNM and not turbulence-boosted	$2.6 \times 10^4\,\mathrm{Myr}$
$T = 10^6\,\mathrm{K}$, $n_{\mathrm{H}} = 10^{-3}\,\mathrm{cm}^{-3}$	$191\,\mathrm{Myr}$	Not a practical dust-growth regime in the Dubois framework	off ($T > 10^4$)

4 Representative Gas Conditions

Table 4 compares the current small-grain growth time, the Dubois reference behaviour, and the proposed revised model for a few representative conditions. For the numerical examples we use

$$Z_{\mathrm{gas}} = 0.02, \qquad D_{\mathrm{tot}} = 0.004, \qquad a_{\mathrm{gr,s}} = 3 \times 10^{-4}\,\mu\mathrm{m}, \qquad f_{\mathrm{ref}} = 0.1.$$

5 Limitations of the 3-Scalar Approximation

The revised model in equation (10) is still only a rough approximation. It does not include: (i) separate silicate and carbonaceous dust reservoirs; (ii) limiting-element chemistry for silicates; (iii) explicit CO suppression for carbonaceous growth; (iv) Coulomb enhancement factors $E_i(a)$; (v) the turbulence-PDF subgrid boost used by Dubois.

Those omissions are exactly why the proposed model is framed as conservative. It is meant to be harder to make unphysically fast, not to replace a full species-resolved treatment. A more faithful long-term model would require additional tracers for at least the major refractory species and their dust phases.

References

[1] Dubois, Y., Rodríguez Montero, F., Guerra, C., Trebitsch, M., Han, S., Beckmann, R., Yi, S. K., Lewis, J., & Jang, J. K. 2024, "Galaxies with grains: unraveling dust evolution and extinction curves with hydrodynamical simulations," *Astronomy & Astrophysics*, 687, A240.

[2] Weingartner, J. C., & Draine, B. T. 2001, "Dust Grain-Size Distributions and Extinction in the Milky Way, Large Magellanic Cloud, and Small Magellanic Cloud," *Astrophysical Journal*, 548, 296.

[3] Draine, B. T., & Li, A. 2007, "Infrared Emission from Interstellar Dust. IV. The Silicate-Graphite-PAH Model in the Post-Spitzer Era," *Astrophysical Journal*, 657, 810.